

Methods of Theoretical Physics

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January 21, 2025



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PY 355

Abstract

This course was taught by Dr. Claudio Chamon during the Spring 2025 semester at Boston University. We used the text Basic Training in Mathematics: A Fitness Program for Science Students by R. Shankar.

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Chapter 1

Sets

Lecture 1: Syllabus Day

Tue 21 Jan 2025 12:30

The professor is recommending ChatGPT as a tutor. This is awesome. Problem sets due Thursdays at 11:59pm.

Prof asked “What are numbers?” and this chick answered “erm uh numebrrs are uh do you want me to give the set theory definition? Erm its difficult to define numbers it requires complex set theory” lmfaol. We will define numbers as an element of an algebraic structure, like a field or group. We take $0 \in \mathbb{N}$. We use the convention of writing $\mathbb{Z}_2$ instead of $\mathbb{Z}/2\mathbb{Z}$.

A cartesian product is a product in **Set**; an object $A \times B$ together with canonical projections that are final in the category $\mathbf{Set}^{A,B}$.